

Supporting Information for Extra Hidden States as Kullback-Leibler Corrections under Emission Misspecification in Hidden Markov Models

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1 Reproducibility

The numerical study uses no external data. All random seeds are fixed in `numerics/scripts/run_review_simulation.py`, with master seed 20260610. The code and generated outputs are available at <https://github.com/AurelienNicosiaULaval/corrective-hmm-states>.

The review simulation compares three fitted models: a Gaussian-emission HMM with $K = 2$, a Gaussian-emission HMM with $K = 3$, and a two-state HMM with two Gaussian mixture components per state. For the replicated study, the sample sizes are $T \in \{500, 1500, 5000\}$ with 200 replications per sample size. The row-level replication file is `numerics/output/review_replications.csv`; the compact summary is `numerics/output/review_summary_by_T.csv`.

2 Mixture-emission EM algorithm

For hidden state $k \in \{1, \dots, K\}$ and mixture component $m \in \{1, \dots, M\}$, the fitted emission density is

$$b_k(y) = \sum_{m=1}^M \rho_{km} \phi(y; \mu_{km}, \sigma_{km}^2), \quad \rho_{km} \geq 0, \quad \sum_{m=1}^M \rho_{km} = 1.$$

The hidden chain has initial probabilities π_k and transition matrix $A = (a_{k\ell})$. Given current parameters, the scaled forward-backward recursion gives the state posterior probabilities

$$\gamma_t(k) = P(X_t = k \mid Y_{1:T})$$

and transition posterior sums

$$\xi(k, \ell) = \sum_{t=2}^T P(X_{t-1} = k, X_t = \ell \mid Y_{1:T}).$$

Within each state, the component responsibility is

$$\tau_t(k, m) = \frac{\gamma_t(k) \rho_{km} \phi(Y_t; \mu_{km}, \sigma_{km}^2)}{\sum_{r=1}^M \rho_{kr} \phi(Y_t; \mu_{kr}, \sigma_{kr}^2)}.$$

The M-step is

$$\begin{aligned} \pi_k^{new} &= \gamma_1(k), & a_{k\ell}^{new} &= \frac{\xi(k, \ell)}{\sum_r \xi(k, r)}, \\ \rho_{km}^{new} &= \frac{\sum_t \tau_t(k, m)}{\sum_t \gamma_t(k)}, & \mu_{km}^{new} &= \frac{\sum_t \tau_t(k, m) Y_t}{\sum_t \tau_t(k, m)}, \end{aligned}$$

Table 1: Selection frequencies over fixed-seed replications.

T	reps	mean Δ_G	$\Pr(\Delta_G < 0)$	$\Pr(\Delta_M < 0)$
500	200	-14.5	0.450	0.955
1500	200	-100.3	0.485	0.970
5000	200	-457.8	0.520	0.995

$$(\sigma_{km}^2)^{new} = \max \left\{ \sigma_{\min}^2, \frac{\sum_t \tau_t(k, m) (Y_t - \mu_{km}^{new})^2}{\sum_t \tau_t(k, m)} \right\}.$$

The implementation uses multiple random starts, scaled likelihoods, a minimum standard deviation $\sigma_{\min} = 0.05$, and sorts states by their fitted state-level means after optimization. For the model used in the paper, $K = 2$ and $M = 2$. The BIC parameter count is

$$(K - 1) + K(K - 1) + K\{(M - 1) + 2M\},$$

corresponding to the initial probabilities, transition probabilities, mixture weights, means, and standard deviations.

3 Replicated selection study

Let

$$\Delta_G = \text{BIC}(\text{Gaussian } K = 3) - \text{BIC}(\text{Gaussian } K = 2),$$

and let

$$\Delta_M = \text{BIC}(\text{Mixture } K = 2, M = 2) - \text{BIC}(\text{Gaussian } K = 3).$$

Thus $\Delta_G < 0$ favours three Gaussian states within the simple Gaussian emission family, whereas $\Delta_M < 0$ favours the enriched two-state mixture-emission model over the three-state Gaussian model.

4 Figures

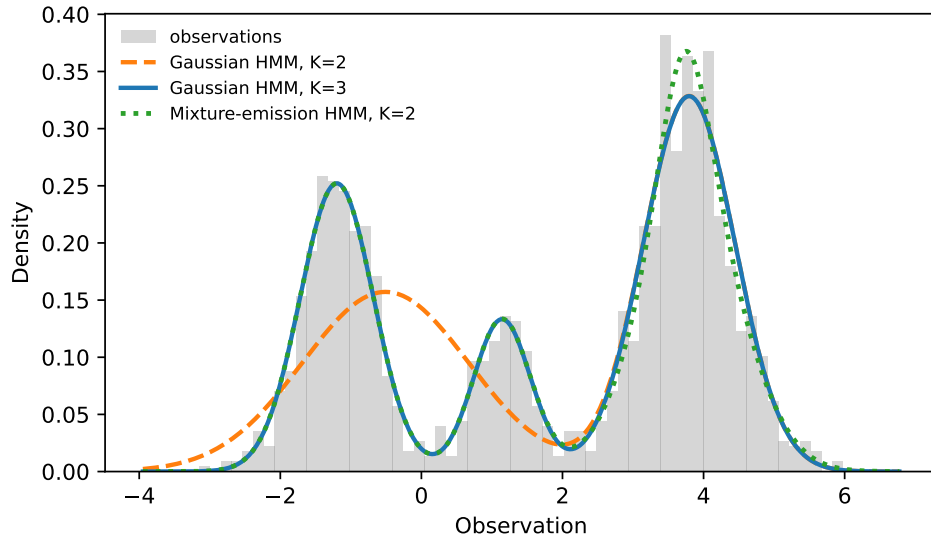


Figure 1: Observed marginal distribution and fitted marginal densities for the reference series.

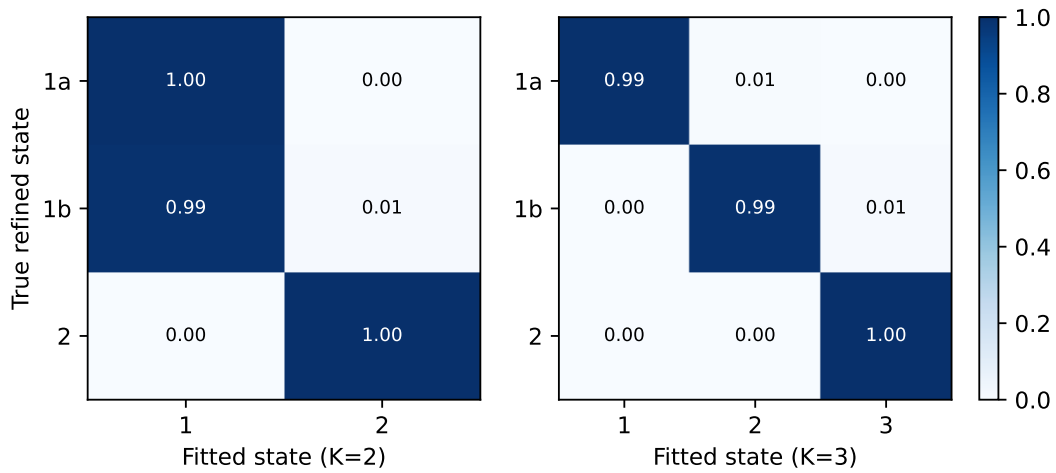


Figure 2: Viterbi decoding proportions for Gaussian-emission HMMs with $K = 2$ and $K = 3$. Rows are true refined states in the simulation.

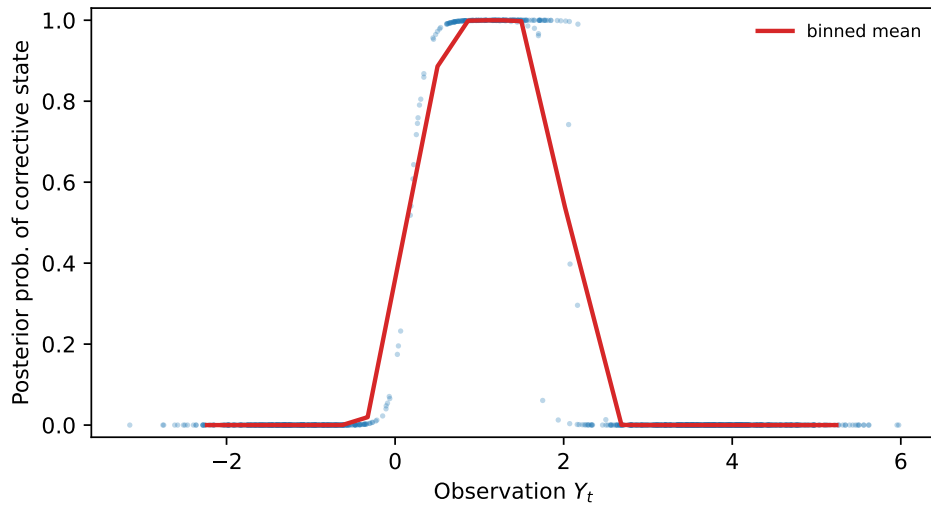


Figure 3: Posterior probability of the corrective Gaussian-HMM state in the reference $K = 3$ fit.

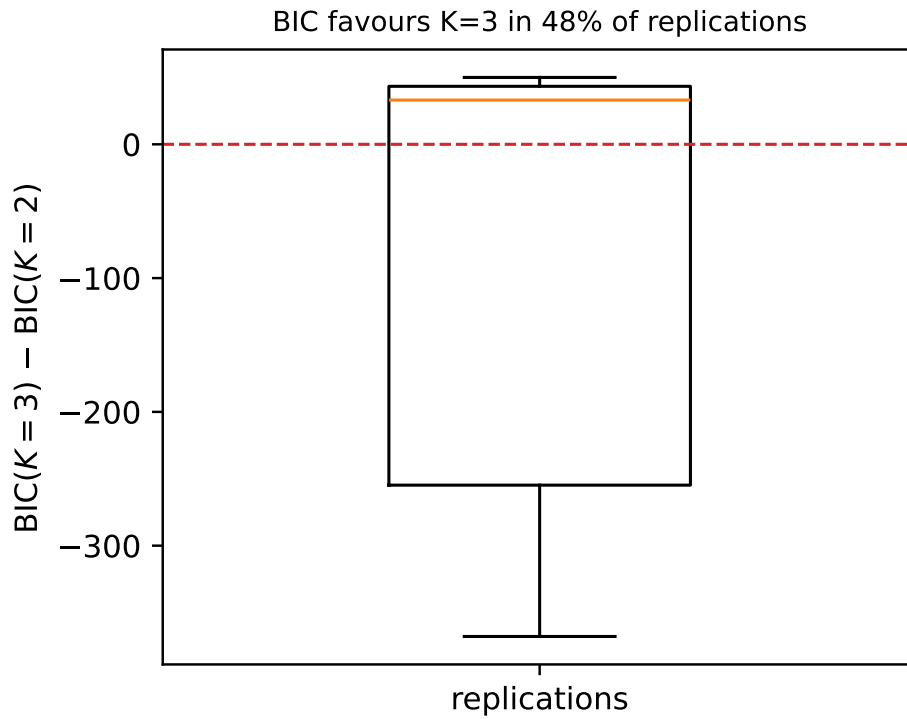


Figure 4: Distribution of Δ_G for the $T = 1500$ replicated study.

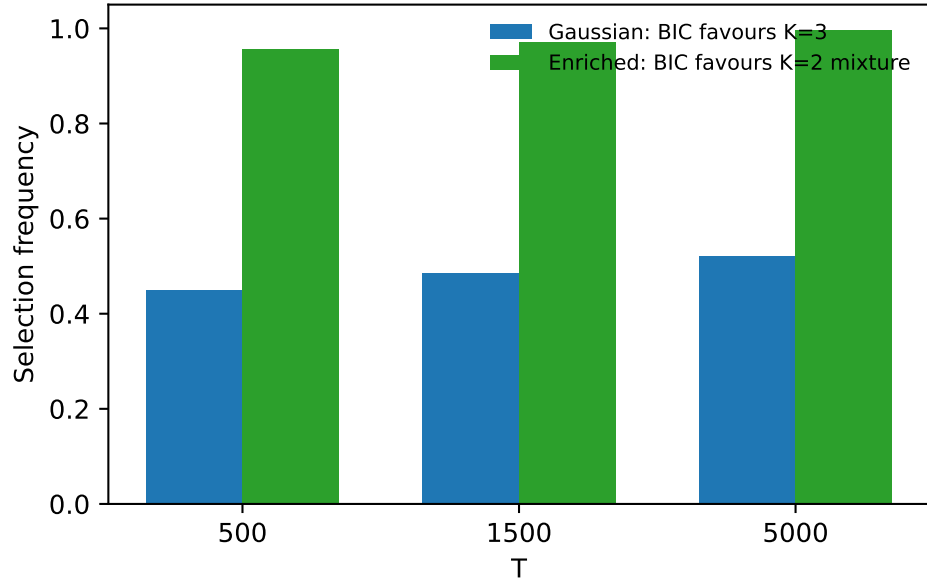


Figure 5: Selection frequencies by sample size for the Gaussian and enriched comparisons.

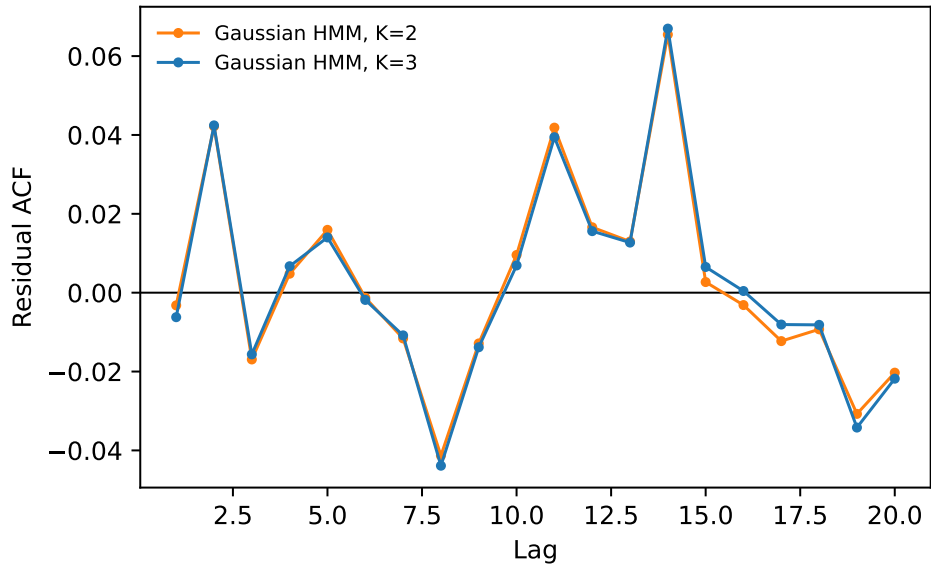


Figure 6: Autocorrelation of approximate one-step standardized prediction residuals for the Gaussian-emission HMMs with $K = 2$ and $K = 3$.